PH3105: Nuclear Physics Laboratory

# Experiment 1


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November 15, 2024

## 1 Aim

To plot and analyse the operational characteristics graph of Geiger Müller (GM) counter. Further we study the distribution curve of the counts from radioactive decay.

## 2 Experimental Setup


**Figure 1:** Experimental setup of the instruments at the laboratory

Our setup consisted of the following apparatus:

**High Voltage Supply** The GM counter requires a high-voltage power supply to operate, typically providing a voltage between 300 and 900 V, depending on the GM tube's specifications. This voltage ensures that the ionization events in the tube result in measurable electrical pulses.

**Na-22 Radioactive Source** A Na-22 source, which emits positrons ($\beta+$ particles) and gamma radiation, is used to simulate the decay process. This radioactive material is placed at a fixed position and distance from the GM counter to measure the number of decays over time.
The Na-22 source undergoes beta decay with the emission of positrons, which, upon annihilation with electrons, produce gamma photons that can also be detected by the GM counter.


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**GM Counter** The GM counter is the primary instrument used to detect ionizing radiation. It consists of a gas-filled tube, typically containing a low-pressure gas (like argon), with a central anode and a cylindrical cathode. When ionizing radiation enters the tube, it ionizes the gas molecules, leading to a cascade of electrons that produces an electrical pulse detectable by the counter. The counter is connected to a counter/display unit that registers the pulses and provides an output (such as counts per minute or counts per second).

**Pulse Counter/Timer** A pulse counter is connected to the GM counter to count the number of pulses (or ionizing events) detected within a specified time period. This allows for the analysis of the counting rate and determination of the operational characteristics.

## 3 Working Principle

### Geiger Müller (GM) Counter

These detectors work by using an electric field to collect and count ion pairs (electrons and positive ions) created when radiation ionizes a gas inside a sealed chamber. The GM counter consists of a cylindrical metallic chamber that serves as the cathode, with a fine wire along its axis as the anode (depicted in figure below). Inert (noble) gases are chosen to fill the counter for their low electron affinity - they do not tend to form negative ions. Argon is most common fill gas: It is inert, stable (non-radioactive), abundant,inexpensive, has reasonably high atomic number and works well with halogen quenching gases (See Fig 2).

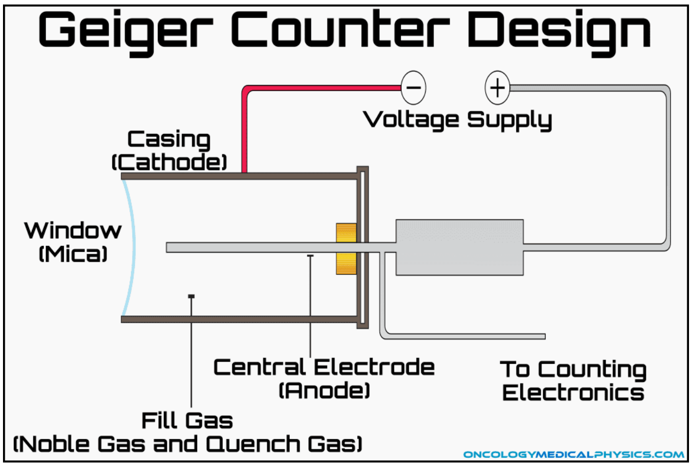



**Figure 2:** Schematic of GM counter

In a cylindrical geometry, electric field($\mathcal{E}$) at a radial distance $r$ from the center is given by

$$\mathcal{E}(r) = \frac{V}{r \ln\left(\frac{b}{a}\right)}$$

where $V$ is voltage applied between anode and cathode, $a$ is anode wire radius, $b$ is cathode inner radius.

This shows that the electric field strength for a given voltage is much larger in cylindrical geometry than in a parallel plate one, which makes the experimental conditions easier to attain. Another advantage of the cylindrical geometry is that *avalanche* (amplification) takes place only in narrow region close to the anode wire which gives good uniformity.

### Townsend Avalanche

The chamber has a sealed window through which radiation like alpha ($\alpha$), beta ($\beta$), or gamma ($\gamma$) particles enter and ionize the gas, typically argon at low pressure.Usually monatomic gas is preferred so that deposited energy is not dissipated in dissociating the molecules. This ionization produces free electrons and positive ions. A high voltage applied across the chamber generates an electric field that accelerates these free electrons toward the anode and the positive ions toward the cathode. This movement of ions creates a current pulse as they reach their respective electrodes. To amplify the initial ionization, the GM counter operates at a high voltage (often over 1000 V), allowing the electrons to gain enough energy to cause secondary ionisations as they collide with other gas atoms in a process called a *Townsend Avalanche*. This avalanche effect exponentially increases the number of ions and electrons, resulting in a strong pulse of current that the counter registers (See Fig 3).

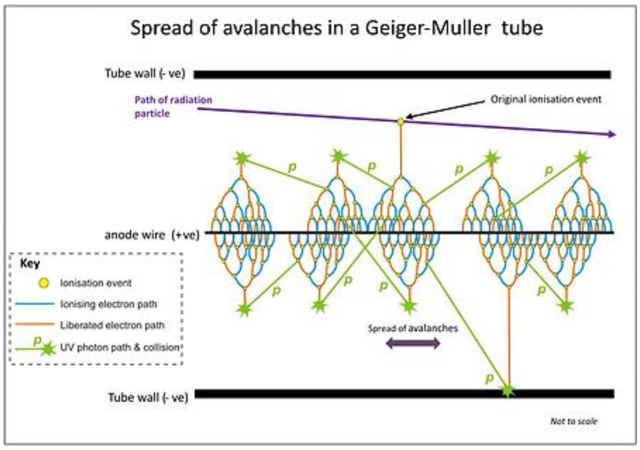


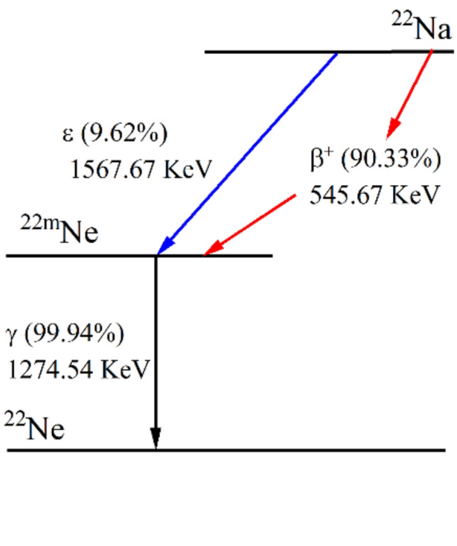

**Figure 3:** Townsend Avalanche

However, without control, the GM tube might experience continuous discharge, making it unresponsive to new ionising events. To prevent this, a *quenching agent*, such as butane, propane, or halogen gases (like chlorine or bromine), is added. The quenching agent absorbs excess energy and neutralises positive ions without triggering further ionization. In organic quenching gases, energy from the positively charged ions is dissipated by breaking molecular bonds, which stops continuous discharge. Once the ions reach the electrodes and are neutralised, the tube resets, ready to detect the next radiation event. If continuous ionization is not prevented the detector will not respond to any other particle except the very first which enters the tube. The GM counter's design makes it suitable for measuring low to moderate levels of radiation, though it cannot provide information about the energy of incoming particles since all pulse sizes are identical regardless of the intensity of the radiation. Due to its simplicity and robustness, the GM counter is popular as a portable radiation detector, though it is limited to applications requiring simple presence detection rather than detailed energy spectroscopy.

## 4 Theory

DECAY SCHEME OF $^{22}_{11}$Na

This radio nuclide undergoes a $\beta^+$ decay with the emission of a positron and a neutrino.



The detailed schematic is as follows:

$$^{22}_{11}\text{Na} \rightarrow ^{22}_{10} {}^{*}\text{Ne} + \beta^{+}(545.67 \text{ keV}) + \nu_e$$

$$^{22*}_{10}\text{Ne} \rightarrow ^{22}_{10} \text{Ne} + \gamma(1274.54 \text{ keV}) + \nu_e$$

In the first decay process has $t_{1/2} \approx 2.6$ yrs; while the de-excitation of the nucleus is almost immediate ($\sim$ 1 pico-second).


**Figure 4:** The Na-22 source used in our experiment

## Law of Large Numbers

The law of large numbers states that the average of the results obtained from a large number of independent random samples converges to the true value, if it exists. More formally, the law states that given a sample of independent and identically distributed (*iid*) values, the sample mean converges to the true mean. In this experiment, since nuclear decay is a spontaneous and random process, the frequency of counts plotted against the number of counts should approximate a Gaussian distribution which will ultimately converge to a poisson distribution at large enough counts.

## Poisson Distribution

A random variable counting the number of occurrences of an event in a specified interval of time is a typical example of a poisson random variable. Each radioactive decay can be assumed to be identical and independently distributed (i.i.d)[1], in which case it is expected to follow poisson distribution. However, one must keep in mind this is a *discrete* probability distribution, characterised by a single parameter $\lambda$. The PMF is given by

$$X \sim \text{Poiss}(\lambda) \implies P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} \quad k \in \mathbb{N}$$

An interesting property of this distribution is that, both the expectation and variance are equal.

$$E[X] = Var(X) = \lambda$$

Now, observe that any random process in nature converges to a gaussian. In this experiment we attempt to find out that if this gaussian also has this special property ($\mu \approx \sigma^2 \rightarrow \lambda$). Then, we can verify our hypothesis that radioactive decay processes indeed follow Poisson distribution at large enough readings.

[1] that the next arrival of pulse(count) is independent from that of the previous arrival of pulse.

# 5 Data Analysis

## Part - 1: Operational Characteristics

| HV (V) | Run 1 | Run 2 | AVG | HV (V) | Run 1 | Run 2 | AVG |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 650 | 138 | 115 | 126.5 |
| 50 | 0 | 0 | 0 | 700 | 126 | 132 | 129 |
| 100 | 0 | 0 | 0 | 750 | 136 | 112 | 124 |
| 150 | 0 | 0 | 0 | 800 | 111 | 124 | 117.5 |
| 200 | 0 | 0 | 0 | 850 | 128 | 124 | 126 |
| 250 | 0 | 0 | 0 | 900 | 136 | 133 | 134.5 |
| 270 | 0 | 0 | 0 | 950 | 116 | 124 | 120 |
| 290 | 0 | 0 | 0 | 1000 | 135 | 106 | 120.5 |
| 300 | 0 | 0 | 0 | 1050 | 123 | 144 | 133.5 |
| 320 | 0 | 0 | 0 | 1100 | 125 | 113 | 119 |
| 340 | 0 | 0 | 0 | 1150 | 162 | 140 | 151 |
| 360 | 0 | 0 | 0 | 1200 | 143 | 103 | 123 |
| 380 | 0 | 0 | 0 | 1250 | 139 | 128 | 133.5 |
| 450 | 0 | 0 | 0 | 1300 | 136 | 140 | 138 |
| 500 | 0 | 0 | 0 | 1350 | 106 | 128 | 117 |
| 520 | 6 | 9 | 7.5 | 1400 | 133 | 136 | 134.5 |
| 550 | 117 | 116 | 116.5 | 1450 | 154 | 126 | 140 |
| 600 | 102 | 123 | 112.5 | 1500 | 139 | 122 | 130.5 |

**Table 1:** Data Readings

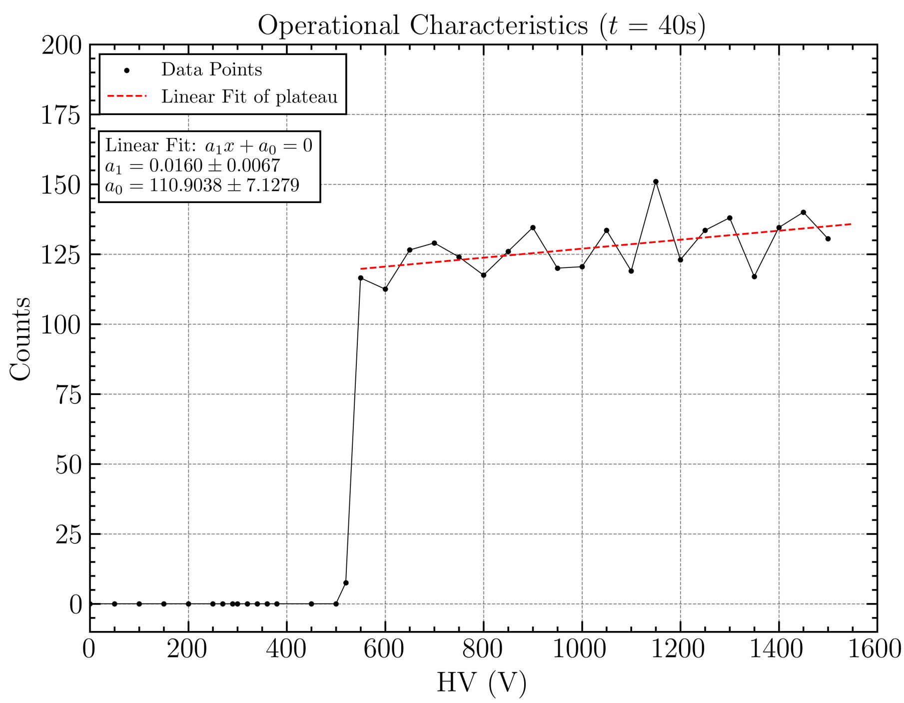



**Figure 5:** Graph of the number of $\beta$ particle counts detected with increasing electrode voltage(V)

Ideally we should get a flat plateau region, but after linear fitting we obtain a non-zero value of the slope (0.0160 ±0.0067) (Fig 5). The reason for which shall be discussed under section 8. We see that the starting voltage lies in 520-550 V range.

**Optimum HV** Optimum HV is generally defined as the middle of the plateau region. In our experiment we have $V_{op} = (600+1500)/2 = 1050$ V. In our next part of the experiment, we shall analyse the radiation

distribution when the GM counter is operated at HV = 1050 V.
However we do NOT find the DC voltage at the **onset of breakdown** because it was out of range of our GM counter.

## Part - 2: Distribution Curve at $V_{op}$ (for $t = 10$s)

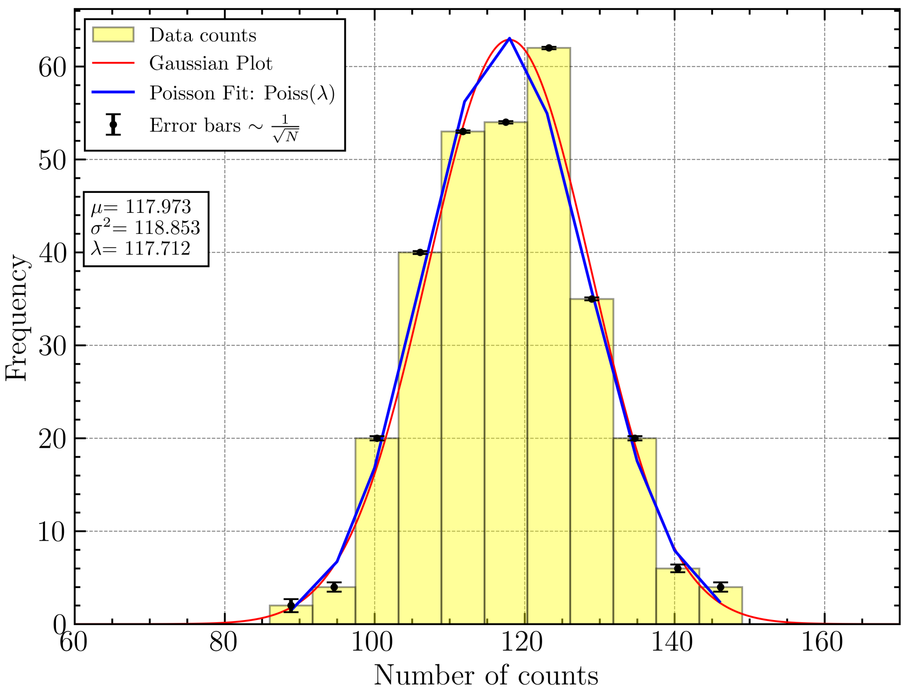



**Figure 6:** Counting Histogram overplotted with a Gaussian of the statistical expectation ($\mu$) and variance $\sigma^2$. Bins = 11, DC Voltage = 1050 V. A Poissonian Fit was also performed at the bin-centers to estimate the value of $\lambda$, to which $\mu$ and $\sigma^2$ converges at large enough readings.

| Counts | Frequency | Counts | Frequency | Counts | Frequency | Counts | Frequency |
|---|---|---|---|---|---|---|---|
| 86 | 1 | 106 | 8 | 120 | 17 | 133 | 2 |
| 91 | 1 | 107 | 8 | 121 | 11 | 134 | 3 |
| 92 | 1 | 108 | 11 | 122 | 13 | 135 | 5 |
| 96 | 1 | 109 | 5 | 123 | 10 | 136 | 3 |
| 97 | 2 | 110 | 6 | 124 | 14 | 137 | 5 |
| 98 | 3 | 111 | 15 | 125 | 7 | 138 | 1 |
| 99 | 3 | 112 | 11 | 126 | 7 | 139 | 2 |
| 100 | 2 | 113 | 6 | 127 | 9 | 140 | 1 |
| 101 | 1 | 114 | 10 | 128 | 9 | 141 | 2 |
| 102 | 6 | 115 | 5 | 129 | 7 | 146 | 2 |
| 103 | 5 | 116 | 9 | 130 | 5 | 148 | 1 |
| 104 | 7 | 117 | 8 | 131 | 5 | 149 | 1 |
| 105 | 6 | 118 | 6 | 132 | 2 | | |

**Table 2:** Counting Statistics Data

To plot the histogram, we have chosen a bin size of 5.72 counts corresponding to a total of 11 bins[2]. We have plotted (not fitted) a gaussian [3] over it to investigate whether our distribution converges to a **Poisson distribution** which is characterized by equal values of mean and standard deviation.

### Gaussian Curve

$$g(x) = A \times N(\mu, \sigma^2)$$

Since we are not plotting a normalised histogram, it is required to scale the amplitude of the Gaussian by a factor of $A$. This is calculated as follows: Let's say we partition our datasets (`counts`) into $k$ bins (which may not be of equal sizes).

Then we have $P = \{\texttt{counts[0]} = x_0, x_1, \ldots x_{k-1}, x_k = \texttt{counts[-1]}\}$. The points in the set $P$ partitioning the dataset `counts` into $k$ bins [4].

Let $f_i$ denote the cumulative frequency in each $i^{th}$ bin (i.e the height of each bar). Then the factor $A$ is simply the total area under histogram.

$$\boxed{A = \sum_{i=0}^{k-1} |x_{i+1} - x_i| f_{i+1}}$$

We took a total of 300 readings, each for $t = 10$ s. The counting statistics is summarized in the following table:

| | |
|---|---|
| $A$ | 1718.181 |
| $\mu$ | 117.973 |
| $\sigma^2$ | 118.853 |
| Median | 119 |
| Highest Count | 149 |
| Lowest Count | 86 |

According to our experiment $|\mu - \sigma^2| = 0.87928 < 1$, indicating a **convergence** towards Poisson distribution. We must also note that the entire readings of Part-2 should be done in the same day, because the detection of counts might depend on the operating conditions of the GM counter such as ambient temperature, gas pressure inside detector etc.

| Bin Center | Total Counts |
|---|---|
| 88.864 | 2 |
| 94.591 | 4 |
| 100.318 | 20 |
| 106.045 | 40 |
| 111.773 | 53 |
| 117.500 | 54 |
| 123.227 | 62 |
| 128.955 | 35 |
| 134.682 | 20 |
| 140.409 | 6 |
| 146.136 | 4 |

**Table 3:** Bin Centers and Total Counts

[2] The optimum bin size was obtained through *hit-and-trial* [3] $N(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}$ [4] `counts[0]` and `counts[-1]` refers to the first and last element of the `counts` dataset (sorted) respectively

## 6 Sources of Error

**Environmental Interference:** The presence of other ongoing experiments in the same room likely caused radiation leakage, which could have contributed to false readings in our setup. This external interference may have led to an overestimation of the count rate.

**Effect of Bin Size on Histogram:** The choice of bin size significantly affected the histogram's appearance, particularly the heights of the bins. While efforts were made to optimize the binning, a different bin size may yield a more accurate distribution.

**Law of Large Numbers** If we would have taken large enough readings, then ultimately $|\mu - \sigma^2| \to 0$, thus implying a convergence towards poisson distribution.

## 7 Conclusion

We were able to detect gamma rays from Na-22 and obtain the operational characteristic curve of the GM counter. However, we were unable to reach the continuous discharge region of the graph, and we postulated that this might be due to the low voltage sensitivity of the GM Counter or the upper limit of High Voltage in this GM Counter. The starting voltage was found in the region (520 V, 550 V). We also performed counting statistics and found the mean of the distribution to be 117.97, and the variance to be 118.85. The histogram was compared with both Gaussian and Poisson distributions, and is successfully found to have a good overlap.

## 8 Discussions

**Shape of distribution** Here the shape of the distribution is observed to have some important consequences. Since it follows a poisson distribution, the variance / spread of the histogram is related to the mean value of observed counts. Which means that, if $\mu$ is low, $\sigma^2$ is also small indicating a more peaked distribution. Conversely, if $\mu$ of counts is much higher, it is more expected to get a broader/wider histogram ($\sigma^2$ large).

**Abrupt jump at HV = 550 V** Below the starting voltage, the electrons produced by ionizing radiation lack enough kinetic energy to cause an avalanche. Instead, they simply drift toward the anode and are collected, producing only a negligible signal. As a result, no events are detected at these low voltages.

Once the voltage exceeds the starting threshold, the electrons gain enough energy to initiate avalanches, leading to the detection of all radiation events. This causes a sudden increase in the counting rate, which is reflected as an abrupt jump in the graph. In our experiment, this starting voltage was found to be around 550 V. Beyond this point, increasing the voltage further does not lead to the detection of additional events as all events produce the same amount of current pulse measured since all events lead to avalanche, resulting in the characteristic flat plateau in the graph.

**Non-zero slope of plateau** In practice, we observe the counting plateau of a Geiger-Müller (GM) tube always has a slight slope rather than being perfectly flat. This slope arises from factors that add low-amplitude signals to the pulse height distribution. One key factor is that certain regions near the tube's ends may have weaker electric fields, leading to smaller discharges when ionization occurs there. Additionally, pulses generated during the recovery time (the period needed for the tube to reset after a discharge) are also smaller than normal.

To minimize the impact of these smaller pulses when measuring the plateau, it's important to keep the counting rate low—specifically, no more than a few percent of the tube's *maximum recovery rate*[5]. For most GM tubes, this means that plateau measurements should be taken at rates below a few hundred counts per second to ensure accurate readings.

[5] refers to the highest rate at which the tube can detect and process individual radiation events. It is determined by the tube's recovery time, which is the period required for the GM tube to return to its initial, ready-to-detect state after registering a pulse and includes its dead time.

Another cause of slope in the plateau of many GM tubes might be due to occasional failure of the quenching mechanism which may lead to a satellite or spurious pulse in addition to the primary Geiger discharge. The probability of such an occurrence is normally low but will increase as the number of positive ions that ultimately reach the cathode also increases. The *active volume*[6] of the tube may also increase slightly with voltage as the very low electric field near corners of the tube is increased and ion pair recombination is thereby prevented.

**Expected onset of discharge** We did not observe the expected continuous discharge at the end of our Geiger-Müller (GM) characteristic curve, even after increasing the applied voltage to the maximum permissible value of 1500 V. Under normal circumstances, such a high voltage should lead to a sharp termination of the plateau region, indicating the onset of continuous discharge. This phenomenon typically arises due to discharges, which are triggered by sharp irregularities on the anode wire, or from multiple pulsing caused by a failure in the quenching mechanism. According to the manual and other groups (who were simultaneously performing the experiment), observed the onset of continuous discharge between 1200-1300 V.

Our inability to detect this behavior could likely be attributed to a reduced voltage sensitivity in our apparatus. This suggests that the actual voltage applied to the GM tube was significantly lower than the nominal voltage indicated, despite setting it to 1500 V. Consequently, the conditions required for the onset of continuous discharge were not met in our setup.

## References


- Knoll, Glenn F. *Radiation detection and measurement.* John Wiley & Sons, 2010.


[6] refers to the specific region inside the tube where ionisation events can occur and be detected. This volume is defined by the space between the anode (usually a thin wire) and the cathode (the tube's outer wall), filled with a gas mixture.